

Human Computer Interaction

ICT3152 - Lecture 09

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Understanding People

Outline

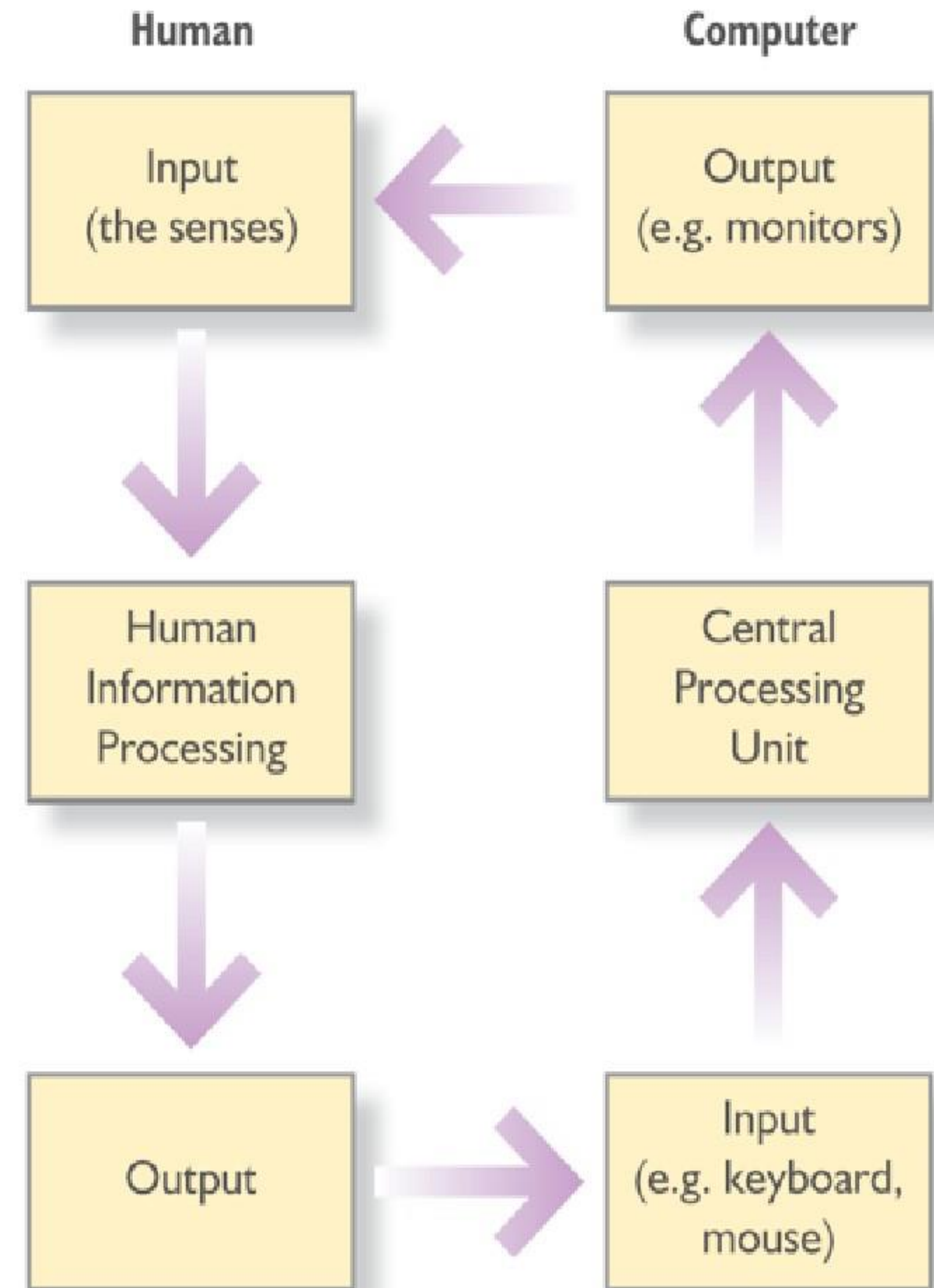
- Cognitive Psychology
- GOMS Model
- Keystroke Level Modeling

Introduction

- Designing usable interactive systems requires an understanding of the capabilities and limitations of people.
- Cognitive Psychology : study of human cognition
- Includes research on:
 - Reasoning
 - memory and attention
 - Perception
 - language
- Research in cognitive psychology has been extremely influential for part of the theoretical foundations of HCI.

Cognitive Psychology and Human Information Processing (HIP)

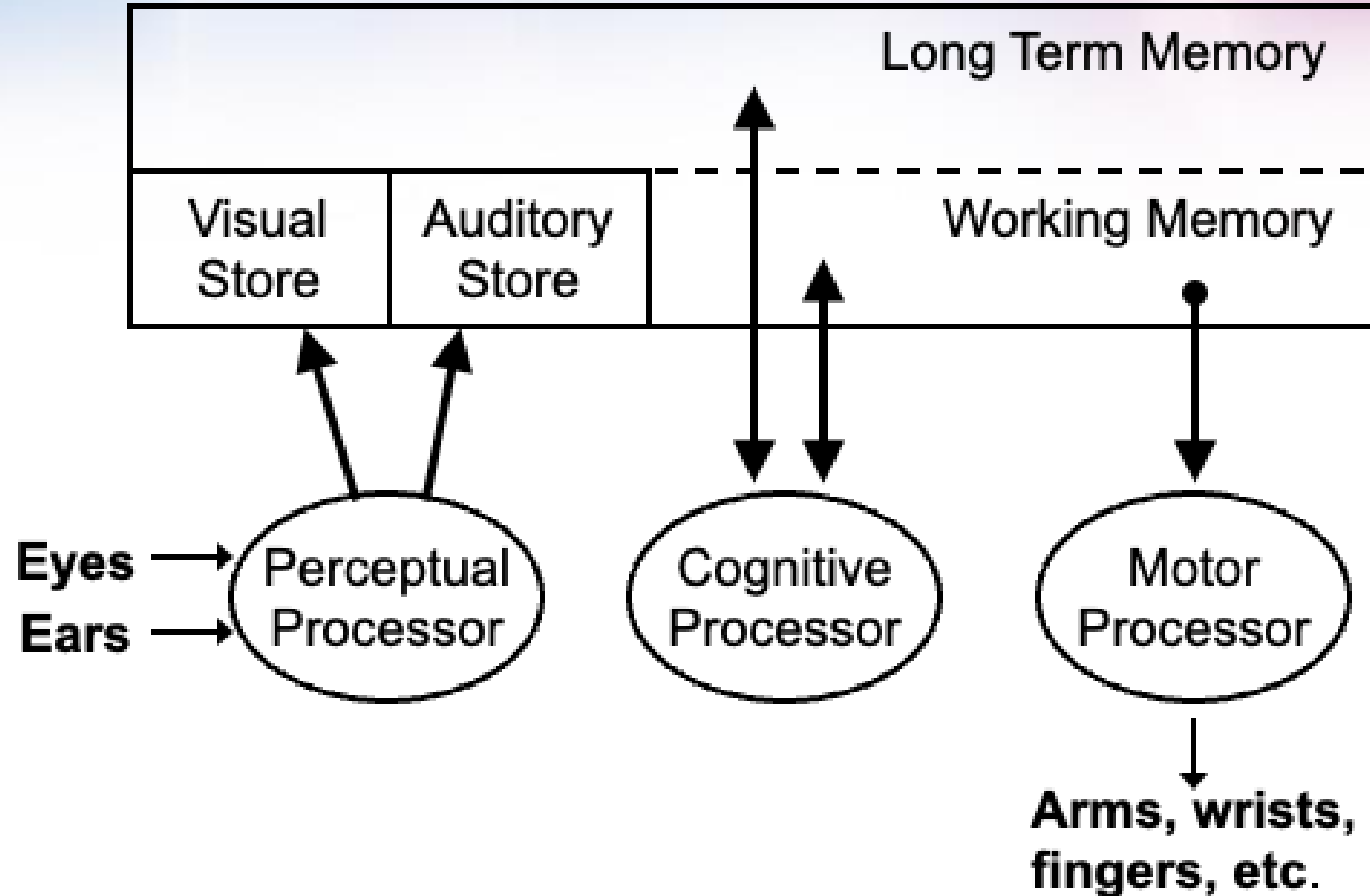
- The Model Human Processor
 - ✓ a simplified model of human information processing
- Three interacting subsystems:
 1. a sensory input subsystem (perceptual subsystem)
 2. a central information processing subsystem (cognitive subsystem)
 3. a motor output subsystem (motor subsystem)
- Human-Computer forms a closed loop:
 - ✓ senses > human information processor > fingers (output)
 - ✓ keyboard input > CPU > output



Model Human Processor

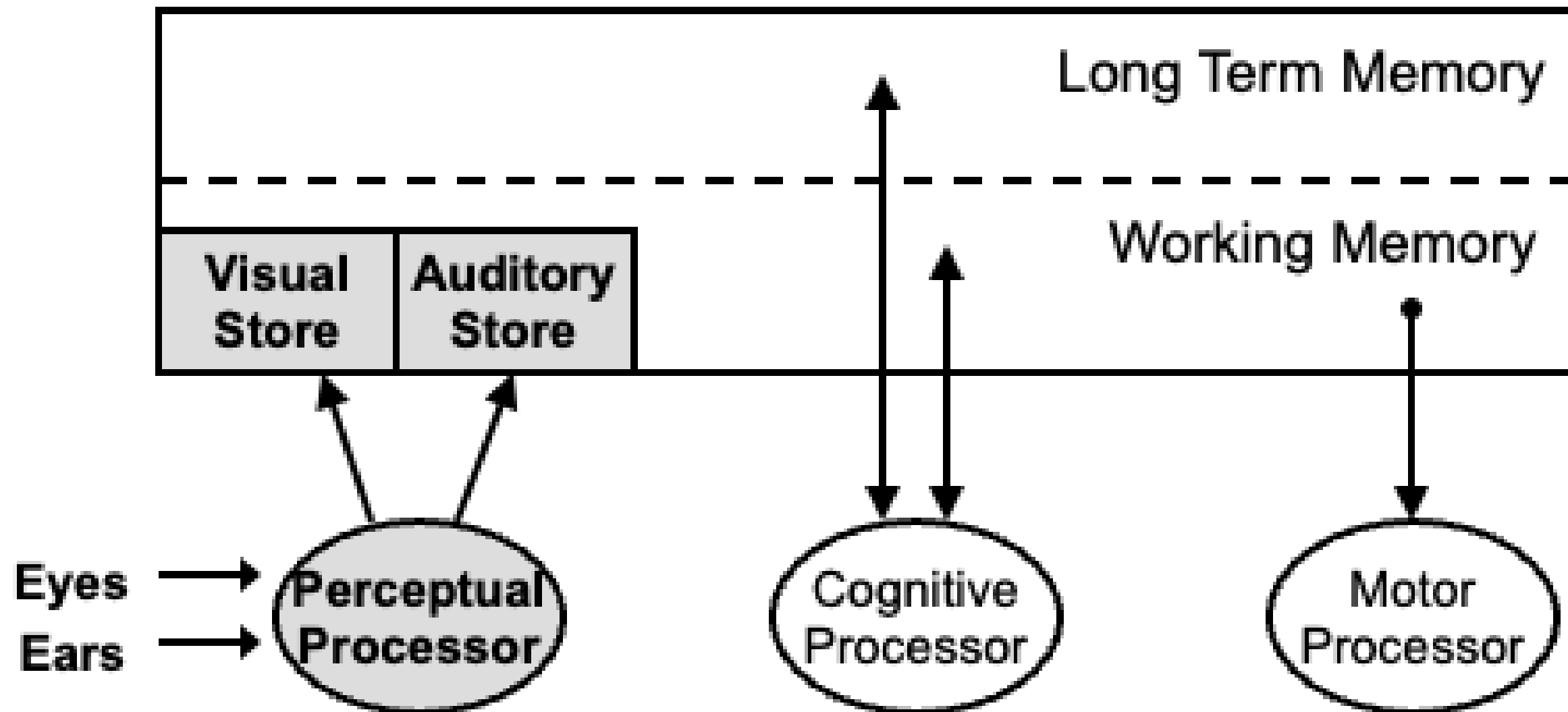
- Contains three interacting systems – Perceptual, Cognitive, and Motor systems
 - For some tasks, systems operate in serial
 - For other tasks, systems operate in parallel
- Each system has its own memory and processor
 - Memory – storage capacity and decay time
 - Processor – cycle time
- Each system guided by principles of operation

Model Human Processor



Perceptual System

- Responsible for transforming external environment into a form that cognitive system can process (form of inputs),
- Composed of perceptual memory and processor



Perceptual Processor

- Shortly after onset of stimulus, representation of stimulus appears in perceptual memory.
 - Representation is physical
 - Eg: ㇿ is just the pattern, not the recognized digit
- As contents of perceptual memory are symbolically coded, they are passed to working memory.
- Decay time
 - 200ms for visual store
 - 1500ms for auditory store

Perceptual Memory

- Codes information in perceptual memory for about 100ms and then retrieved next stimulus.
- Cycle time = $\sim 100\text{ms}$
- Processor cannot code all information before the next stimulus arrives.
- Can utilize information about perceptual system to improve and better understand HCI.

Exercise

Assume perceptual cycle time = 100ms

- 1) If 20 clicks per second are played for 5 seconds, about how many clicks could a person hear?
- 2) If 30 clicks per second are played for 5 seconds, about how many clicks could a person hear?

Exercise

A perceptual cycle time = 100 ms
= 0.1 seconds

✓ Part 1: 20 clicks/second for 5 seconds

- Total clicks played:

$$20 \times 5 = 100 \text{ clicks}$$

- Number of perceptual cycles in 5 seconds:

$$5 \div 0.1 = 50 \text{ perceptions}$$

- Answer: A person could hear about 50 clicks

✓ Part 2: 30 clicks/second for 5 seconds

- Total clicks played:

$$30 \times 5 = 150 \text{ clicks}$$

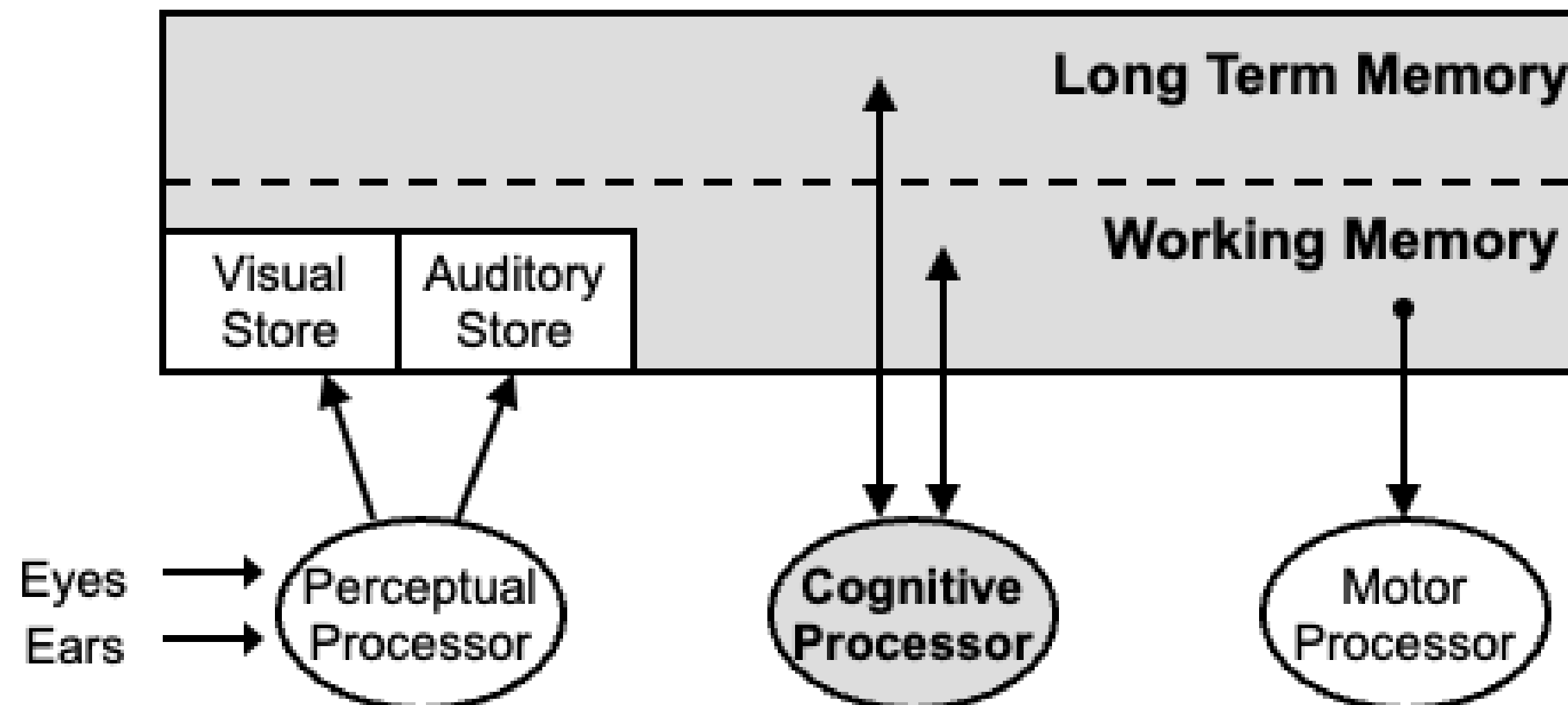
- Same number of perceptual cycles in 5 seconds:

$$5 \div 0.1 = 50 \text{ perceptions}$$

- Answer: A person could still hear about 50 clicks

Cognitive System

- Uses contents of working memory and long-term memory to make decisions and schedule actions with motor system.
- Composed of a processor and two memories.
 - Working memory and long-term memory



Working Memory

- Holds intermediate products of thinking and representations produced by perceptual system.
- Comprised of activated sections of long-term memory called “chunks”
 - A chunk is a hierarchical symbol structure
 - 7 ± 2 chunks active at any given time

Long-Term Memory

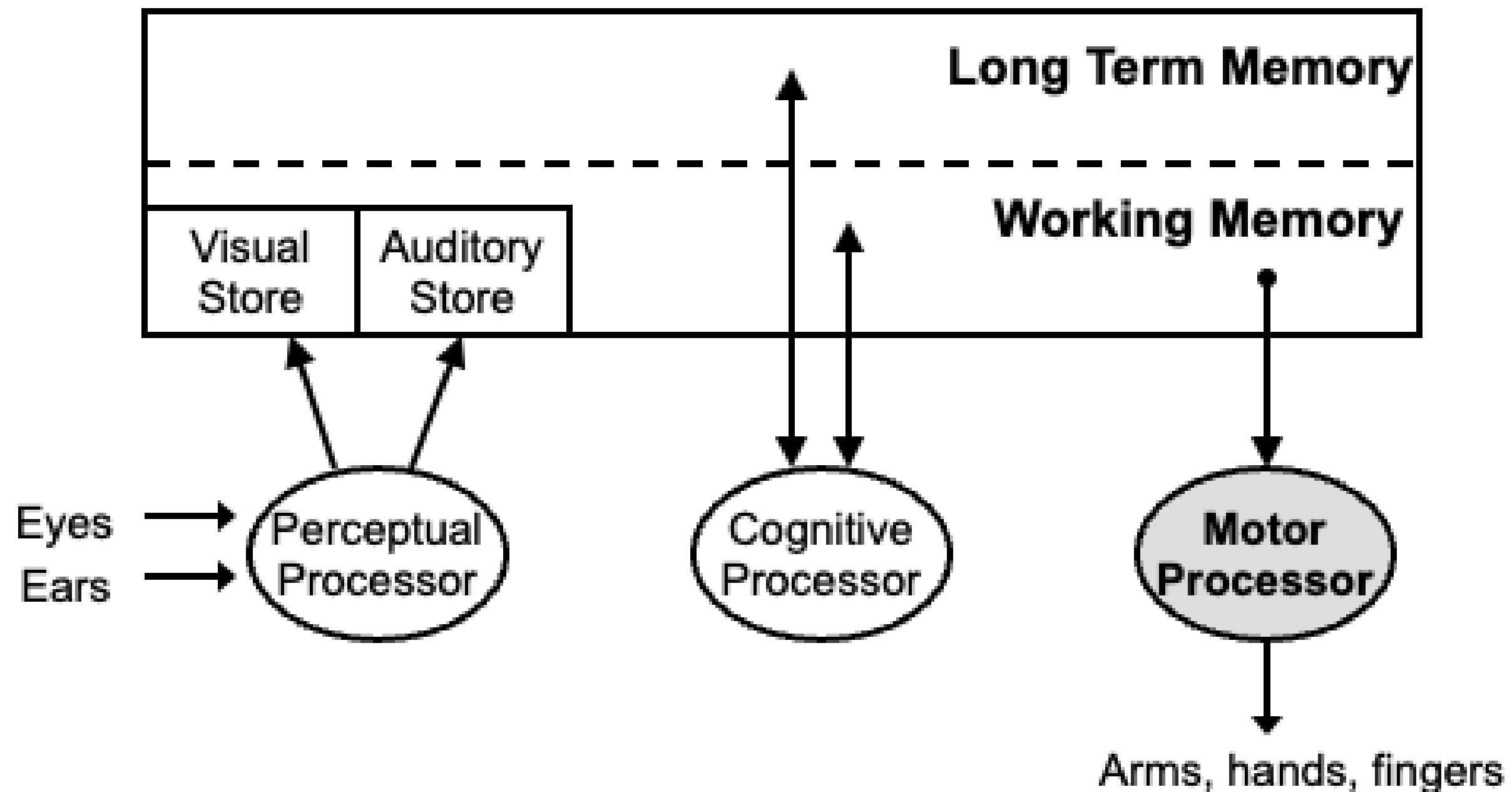
- Holds mass of knowledge; facts, procedures, history.
- Consists of a network of related chunks where edge in the network is an association (semantic network).
- Fast read, slow write.
- Infinite storage capacity, but you may forget because:
 - Cannot find effective retrieval cues.
 - Similar associations to other chunks interfere with retrieval of the target chunk (discrimination principle).

Cognitive Processor

- Based on recognized-act cycle
 - Recognize - activate associatively-linked chances in long term memory.
 - Act - modify contents of working memory.
 - Cycle time = $\sim 70\text{ms}$

Motor System

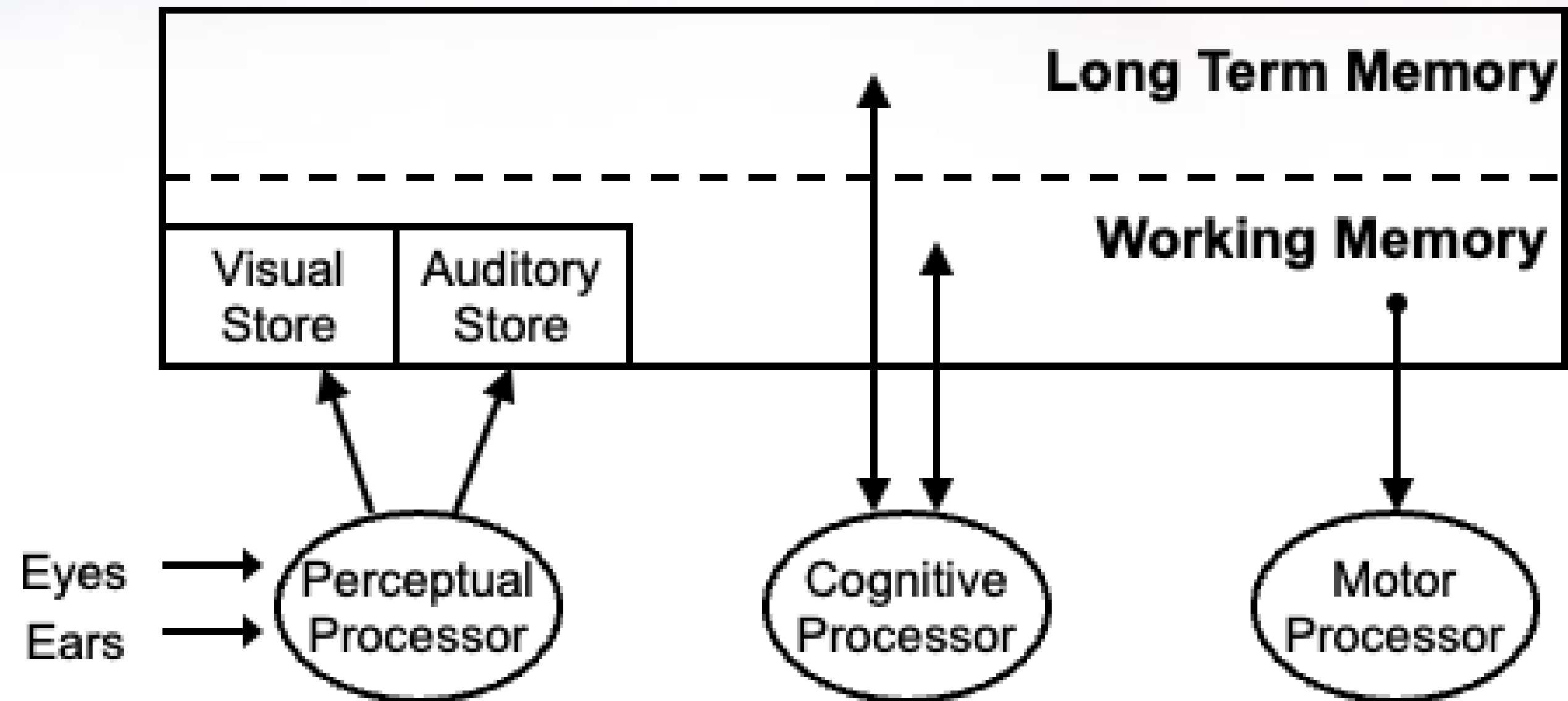
- Translates thoughts to actions (form outputs)
 - Head-neck, and arm-hand-finger actions



Motor Processor

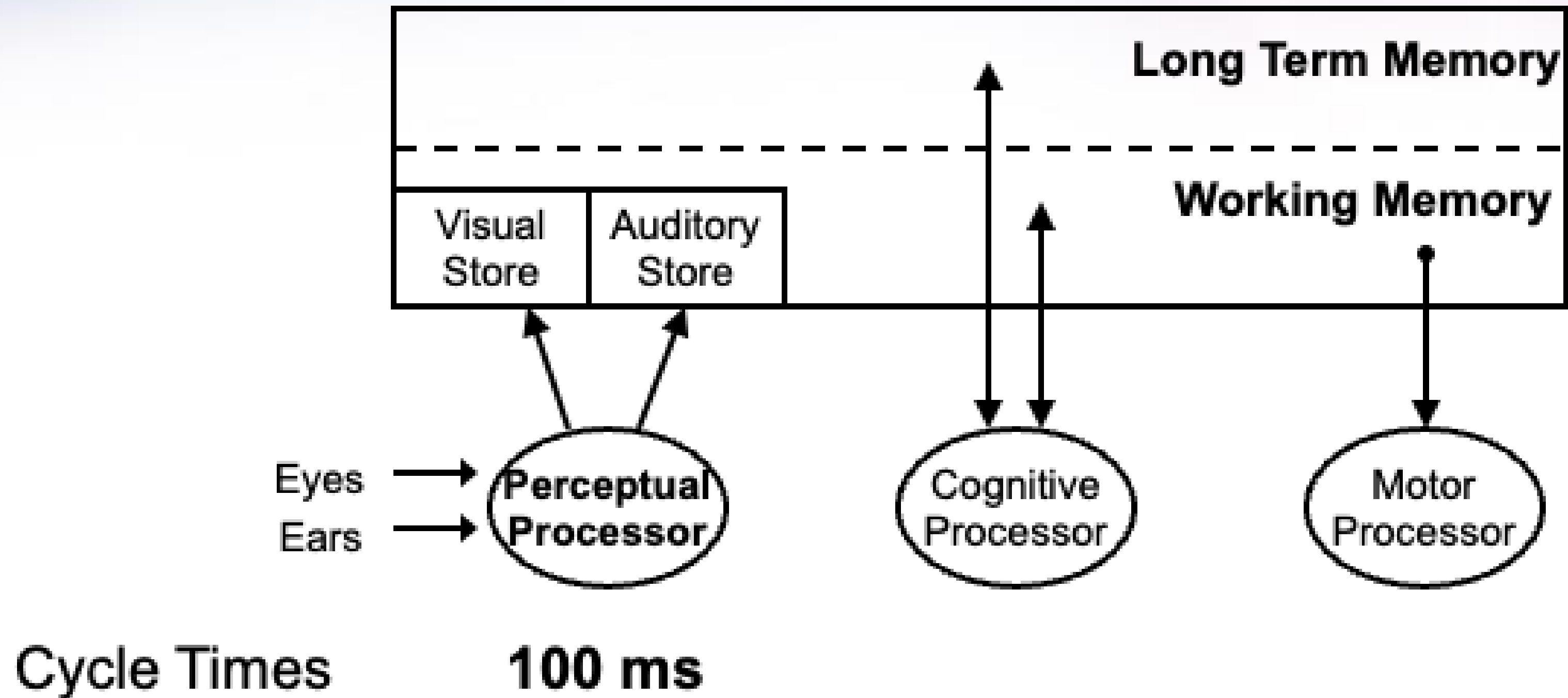
- Controls movement of body.
 - Movement composed of discrete micro-movements.
 - Micro-movement lasts about 70ms.
 - Cycle time of motor processor about 70ms.
- Caches common behavior acts such as typing and speaking.

What we know so far?

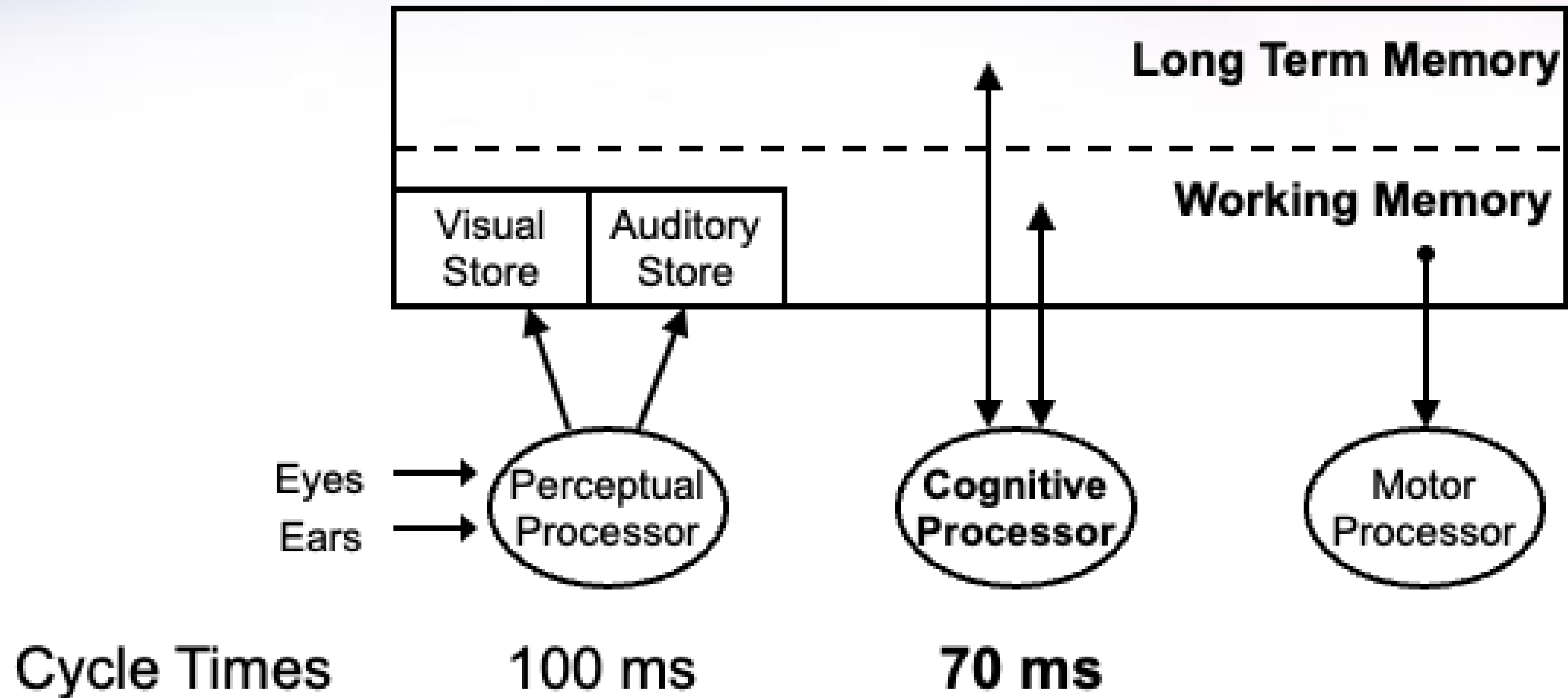


Cycle Times

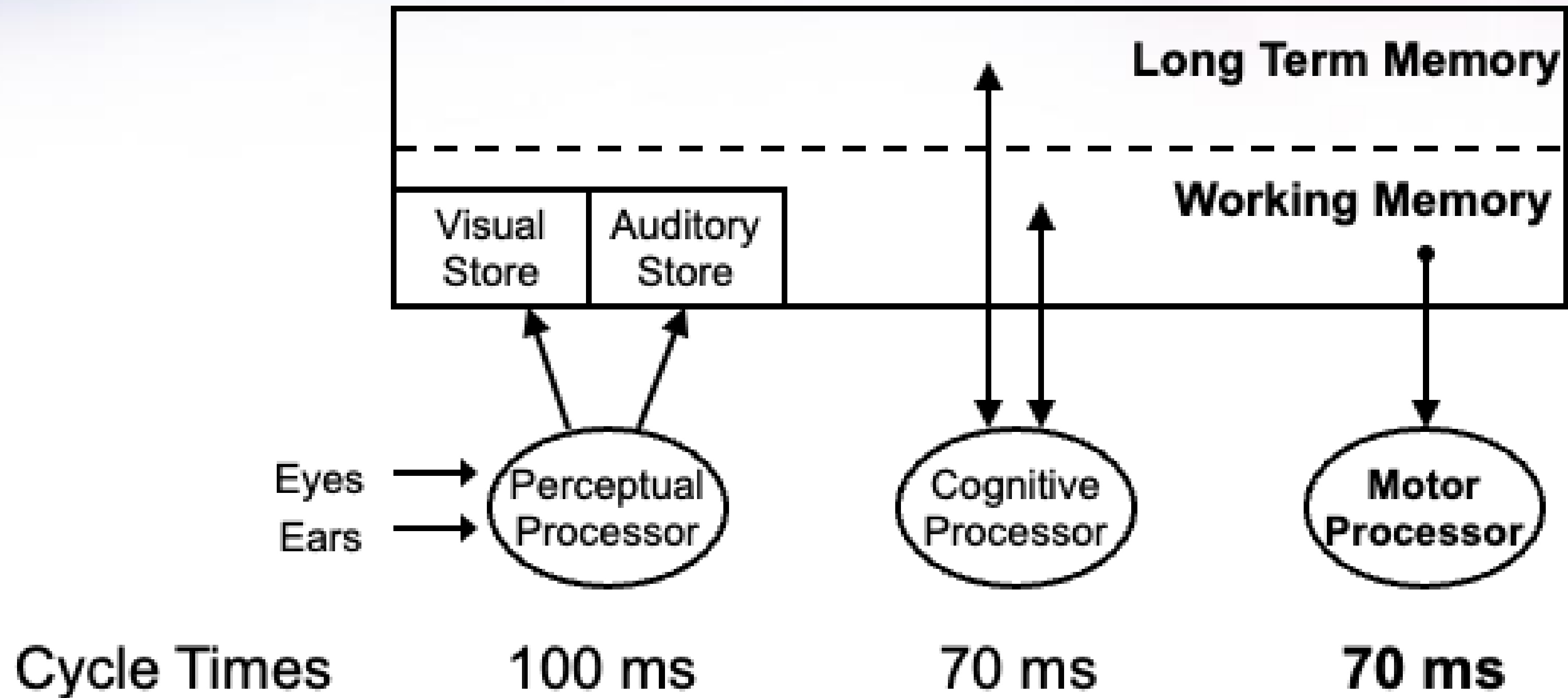
What we know so far?



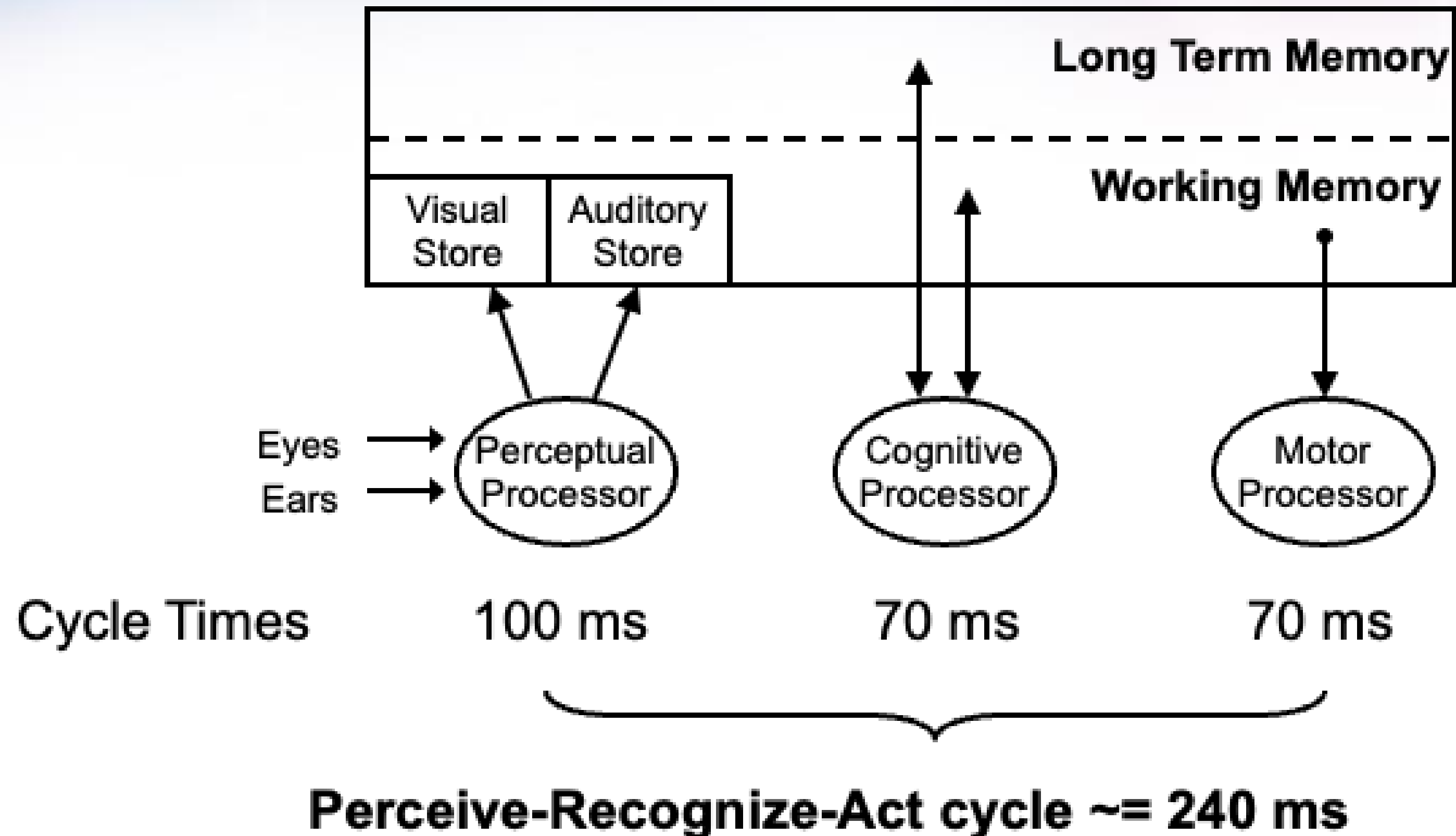
What we know so far?



What we know so far?



Model Human Processor



Use Model to Compute Reaction Time for Simple Matching Task

- A user sits before a computer terminal. Whenever a symbol appears, s/he must press the space bar. What is the time between stimulus and response?

Use Model to Compute Reaction Time for Simple Matching Task

- A user sits before a computer terminal. Whenever a symbol appears, s/he must press the space bar. What is the time between stimulus and response?

$$T_p + T_c + T_m = 240 \text{ ms}$$

Use Model to Compute Reaction Time for a Symbol Matching Task

- Two symbols appear on the computer terminal. If the second symbol matches the first, the user presses “Y” and presses “N” otherwise. What is the time between the second signal and response?

Use Model to Compute Reaction Time for a Symbol Matching Task

- Two symbols appear on the computer terminal. If the second symbol matches the first, the user presses “Y” and presses “N” otherwise. What is the time between the second signal and response?

$$T_p + 2T_c(\text{compare} + \text{decide}) + T_m = 310 \text{ ms}$$

In General Case

- Analyze goal structure of the task, then for each step:
 - Analyze user actions required (motor system)
 - Analyze user perception of the output (perceptual system)
 - Analyze mental steps to move from perception to action (cognitive system)
- Sum the processing times from each step to get a reasonably accurate prediction of task performance.

GOMS Model

- Models task structure (goals) and user actions (operations, methods, selection rules)
 - ✓ **G**oals – cognitive structure of a task
 - ✓ **O**perators – elementary acts that change user state or task environment
 - ✓ **M**ethods – set of goal-operator sequences to accomplish a sub-goal
 - ✓ **S**election – rules to select a method
- Assumes error free and rational behavior

Goals

- Some tasks that the user tries to accomplish.
- Typically described in an action-object pair in the form <verb noun>
- A goal can be defined at any level of abstraction.
- Examples are
 - Delete word
 - Move-left cursor
 - Insert database-entry
 - Compose text
 - Delete file

Operators

- The actual, basic actions the user executes like goals, they are also represented as an action-object pair.
- The difference between goals and operators is that the goal is something the user wishes to achieve, while the operator is just something the user executes.

Goal	Operator
Move-left cursor	Press backarrow

Methods

- Sequence of steps to accomplish a goal
 - goal decomposition
 - can include other goals
- Assumes method is learned & routine
- Examples
 - drag file to trash
 - retrieve from long-term memory command
 - restore

Selection Rules

- If there are more than one method to accomplish a goal, Selection rules pick method to use.
- Examples
 - IF <condition> THEN accomplish <GOAL>
 - IF <hand is on the key board> THEN <press relevant key>
 - IF <hand is on the mouse> THEN <click relevant mouse button>

Example - Online Dictionary Lookup

- Goal - retrieve definition of a word
 - Goal - access online dictionary
 - Operator - Type URL sequence
 - Operator - Press Enter
 - Goal - lookup definition
 - Operator - type word in entry field
 - Goal - submit the word
 - Operator - move cursor from field to lookup button
 - Operator - select lookup
 - Operator - read output

GOMS Model - Variations

- **Keystroke Level Model (KLM)**: The first, simplest form of GOMS consisting of the sum of subtasks and required overhead.
- **Card Moran Newell (CMN)–GOMS**: A serial stage model of GOMS.
- **Critical Path Method (also known as Cognitive Perceptual Motor or CPM–GOMS)**: A parallel stage model (for users with highest level of skill) critical-path-method or cognitive-perceptual motor analysis of activity – perceptual, cognitive, motor operators can be performed in parallel as the task demands.

Keystroke Level Modeling (KLM)

- Simple tool that allows to predict or estimate how long it will take an experienced user to complete a routine task in their software.
- The model is composed of six elements or operators:
 - K – Keystroke or button press
 - P – Pointing with a mouse
 - H – Homing the hands on the keyboard or other device
 - D – Manually drawing
 - M – Mental preparation
 - R – System response time

KLM Parameters

Operator	Abbrev.	Description	Duration in sec.
<i>Physical</i>	K	Keystroke or a button press	
		• Best typist	0.08
		• Average non-secretary typist	0.28
		• Worst typist	1.20
	P	Pointing with a mouse to a target element on a screen	1.10
	H	Homing hands to keyboard or mouse	0.40
<i>Mental</i>	M	Estimated mental operator for preparing to execute a physical action	1.35
<i>Response</i>	R	System's response time (t) measured in seconds	
		(User's wait time until system responds)	t

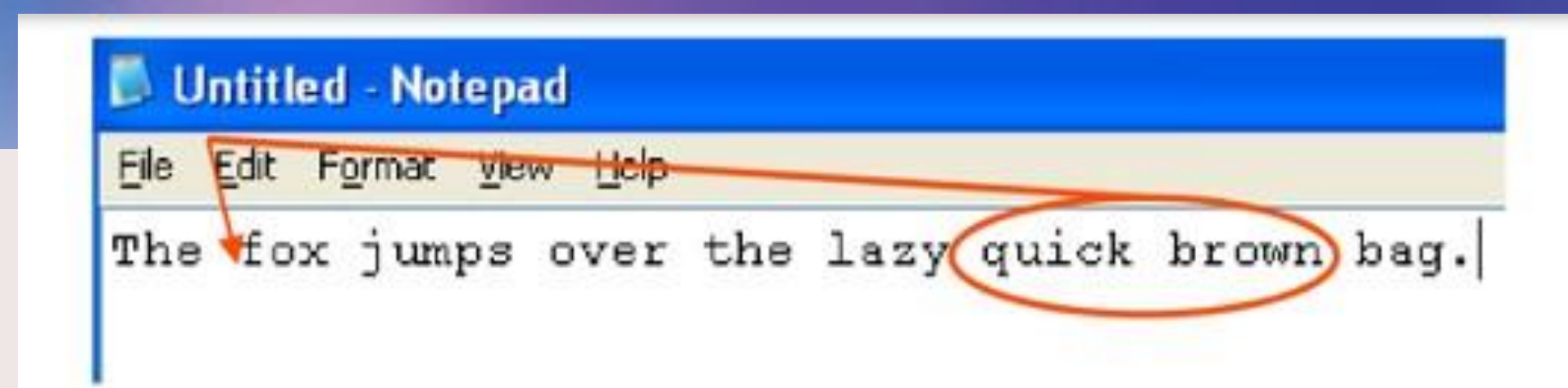
Example

- Enter a street address into a text field
 1. Initiate the entry (M)
 2. Find the correct text field (M)
 3. Point to the correct field (P)
 4. Press mouse button (K)
 5. Release mouse button (K)
 6. Move hands from mouse to keyboard (H)
 7. Type “123 Main St.” (14K) – It’s key press, not characters.
- Total Time = $2M + 1P + 1H + 16K$

Approach

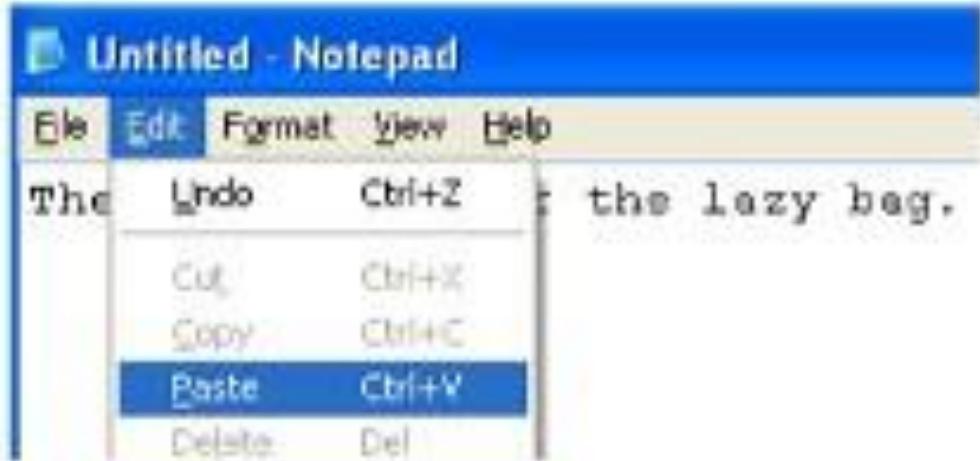
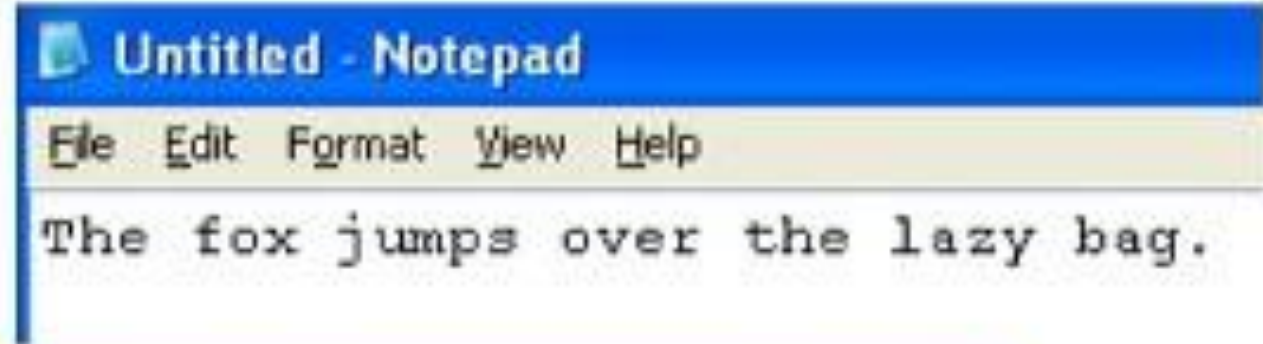
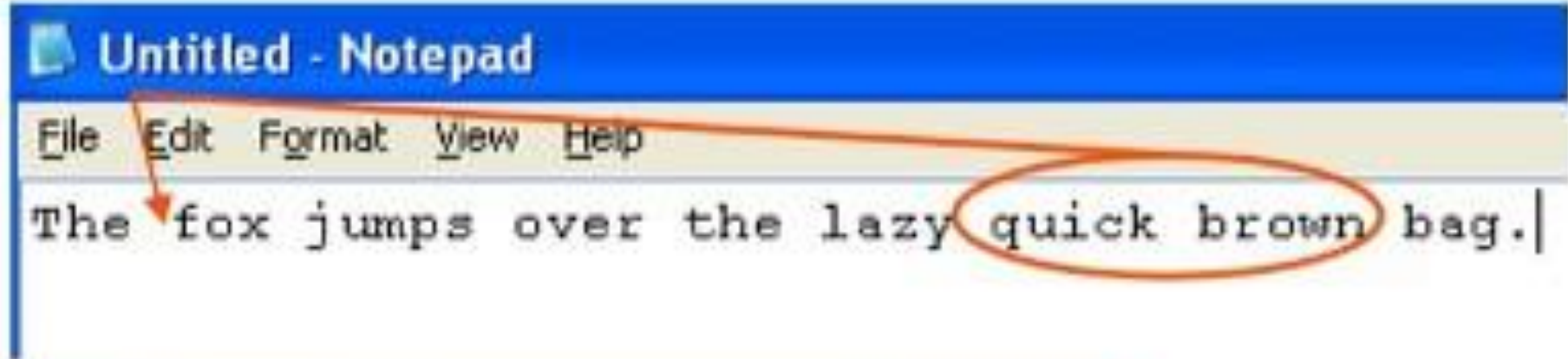
- How to apply the KLM to estimate the execution time required by a specified interface design:
 1. Choose one or more representative task scenarios.
 2. Have the design specified to the point that keystroke-level actions can be listed for the specific task scenarios.
 3. For each task scenario, figure out the best way to do the task, or the way that you assume users will do it
 4. List the keystroke-level actions and the corresponding physical operators involved in doing the task.
 5. If necessary, include operators for when the user must wait for the system to respond.
 6. Insert mental operators for when user has to stop and think.
 7. Look up the standard execution time to each operator.
 8. Add up the execution times for the operators.
 9. The total of the operator times is the estimated time to complete the task.

Example (Using GOMS)



Top Level Goal	Edit Manuscript (move “quick brown” to before “fox”)
Sub goal	Highlight text
Operators	Click mouse-button
	Move-mouse
	Type characters (keyboard shortcuts)
Methods	1. Delete-word-and-retype (retype method)
	2. Cut-and-paste-using-keyboard-shortcuts (shortcuts method)
	3. Cut-and-paste-using menus (menus method)
Selection Rules	If the text to be moved is one or two characters long, use retype method
	Else, if remember shortcuts, use shortcuts method

Example -
Cut-Paste
with Menus
(using
KLM)



M=1.35
P=1.10
K=0.20

Description	Operator
Mentally Prepare	M
Move cursor to "quick"	P
Double-click mouse button	K
Move cursor to "brown"	P
Shift-click mouse button	K
Mentally Prepare	M
Move cursor to Edit Menu	P
Click mouse button	K
Move cursor to Cut menu item	P
Click mouse button	K
Mentally Prepare	M
Move cursor to before "fox"	P
Click mouse button	K
Mentally Prepare	M
Move cursor to Edit menu	P
Click mouse button	K
Move cursor to Paste menu item	P
Click mouse button	K
TOTAL PREDICTED TIME	



Thank
you!